addac112 FRIPPER

This is the addac112 looper firmware with a second engine for the left knob section: **TAPE**, an emulation of Frippertronics. Two tape machines sit side by side, the tape running from the record head of the first to the playback head of the second, and the playback is fed back into the record input.

It was built from the 112 dual firmware (`addac112_cdskip/112_cdskip`) with its CD SKIP engine replaced by TAPE. The looper, banks, SD card, presets, loop pitch, sync and REC prob/delay are the same in both modes.

```
engine/     tape_engine.h (DSP), tape_param_map.h (knob curves); no firmware dependencies
test/       desktop tests of the engine: ./run_tests.sh
firmware/   the 112 firmware (src/, web/ editor, Makefile)
```

```
cd firmware
make                # build/FRIPPER_<version>.bin
make program-app    # bootloader in DFU mode
```

It builds against addac112's libdaisy / DaisySP / CMSIS (`make ADDAC112_DIR=...` to override). libdaisy is linked at `-O2` (`build_libdaisy.sh` builds it into `libdaisy_O2/`, and `make` runs it when it's missing; the vendored build is `-O0`). The tape engine (`tape_bridge.cpp`) and the looper (`granular.cpp`, `loop.cpp`) are built with `-O2`, and the looper's per-sample code and libm run from ITCM (`STM32H750IB_sram_itcm.lds`). The audio callback is timed in CPU cycles, because `hw.GetUs()` wraps every ~17.9 s.

Switching engines

Each engine is its own firmware. This one always runs TAPE. Plain granular is the stock addac112 firmware.

1. Put the firmware `.bin` files (112, CD SKIP, FRIPPER, ...) in a folder named `ENGINES` on the SD card.
2. Go to **Menu** → **ENGINE** (the last item of the main menu). It lists the `.bin` files in `ENGINES`.
3. Choose one and confirm **FLASH & RESTART? YES**.

- The module programs the chosen firmware and restarts into it. The `.bin` stays in `ENGINES`, so you can switch back the same way. The bank selection and the calibration are kept.
- `.bin` files in the SD root are still installed by the bootloader at power-up, as before. The bootloader ignores the `ENGINES` folder.
- The file is checked before anything is erased. If the SD card fails during the write, the module stays on and asks you to flash again. Do this before powering off. If that is not possible, put a `.bin` in the SD root and power-cycle.
- The old `engine_mode=` key in `GLOBAL.CFG` is ignored.

Presets store both engines' values (the stock grain keys plus `tape_*` keys). The stock 112 firmware ignores the `tape_*` keys, but if it saves a preset, those keys are dropped from that preset.

TAPE controls (dual-label overlay)



The engine's labels are in orange; the rest of the panel works as on the stock 112.

Row	Col 1	Switch	Col 2	Col 3
1	POSITION / **GAP**	START GRID / **GAP SYNC** (FREE / ÷8 / ÷4)	POS DEV / **WOW**	GRAINS ACTIVE / **HEADS** (jack in trigger mode: **TAP**)
2	PLAY LENGTH / **REGEN**	SIZE GRID / **SPEED RANGE** (3¾ / 7½ / 15 ips)	LENGTH DEV / **SATURATE**	ATTACK / **SWELL**
3	GRAIN DELAY / **SPACING**	SKIP CLOCK / **STEREO** (LINK / DRIFT / PING)	DELAY DEV / **AGE**	DECAY / **TONE**
4	DIRECTION / **REVERSE**	REPEAT / **HOLD**	PROB/REPEAT / **DROPOUT**	GRAIN PITCH / **VARISPEED** (+1V/oct)
5	VOLUME MIN / **DUCK**	–	VOLUME DEV / **FLUTTER**	PANNING / **WIDTH**

Knobs

- **GAP** sets the distance from the record head to the main playback head, as a delay at 7½ ips. The range is 50 ms to 40 s (about 20 s at 96 kHz) and the centre is 4 s. Turning it moves the heads along the tape, so the pitch bends while they move (GAP MODE: GLIDE / JUMP).
- **REGEN** is the feedback. 100 % is at about 91 % of the knob, and above that the repeats grow into **SATURATE**.
- **HEADS** sets 1–4 playback heads. The extra heads sit between the record head and the main head: at even fractions of the gap when **SPACING** is at 0, at unrelated ratios when it is at 100 %.
- **REVERSE** turns heads around, the main head last.
- **AGE** is the loss on every pass: lows, highs, and hiss.
- **DROPOUT** is worn oxide: short dips of all playback heads at once.
  - How often: up to about 3 per second at full (it grows with the square of the knob, so about one every 1.3 s at the centre).
  - Each dip lasts 10–120 ms and goes down to 0–40 % of the level, with 5 ms edges.

    - The dips happen before the feedback, so each hole is recorded and comes round again on later passes, fading with REGEN. With HOLD on, the holes stay.
- **DUCK** lowers the tape output while you play, so live notes sit on top of the loop.
    - It listens to what goes on the tape (before INPUT LEVEL and SWELL). With TAPE INPUT at LOOP or BOTH, the loop playback ducks the tape too.
    - It starts at about –50 dBFS and reaches the knob's depth at about –20 dBFS. It reacts in about 5 ms and recovers over about 300 ms.
    - Only what you hear is ducked. The feedback and what is recorded stay at full level, so the loop keeps building underneath. The dry signal and MONITOR aren't ducked.
- **SWELL** is an automatic volume pedal: every new note fades in on the tape (4 ms look-ahead).
    - With SWELL TIME at **GAP** (the default), the knob is the fade-in as a share of the loop period: 50 % on a 4 s loop is a 2 s swell. It follows GAP, SPEED RANGE and VARISPEED, within 30 ms to 8 s. The readout shows both, e.g. `50 % GAP (2.00 s)`.
    - At **FIXED** it is 30 ms to about 2.5 s, whatever the gap.
- **VARISPEED** is the tape speed on top of SPEED RANGE (–24…+36 st, snapped to the QUANTIZE scale). As on a real tape, what is already on the tape changes pitch, while what is recorded at the new speed comes back at its own pitch.

Switches

- **GAP SYNC** snaps the gap to multiples of a ÷8 or ÷4 slice of the selected loop. Octave steps of SPEED RANGE keep it on the grid.
- **SPEED RANGE** is ×0.5 / ×1 / ×2 (an octave down or up, with twice or half the delay). The motor ramps between speeds.
- **STEREO**:
    - **LINK**: both channels at the same gap.
    - **DRIFT**: the right channel's heads sit 2.9 % further away, so the two sides slowly phase.
    - **PING**: the input goes to the left tape and each pass changes side.
- **HOLD** holds REGEN at 100 % with no AGE, so nothing fades. HOLD INPUT chooses whether the input keeps recording on top (sound on sound) or the loop freezes.

Other panel controls in TAPE mode

- **GRAINS VOLUME** is the tape level.
- **GRAINS FEEDBACK** records the tape into the loop: print a Frippertronics bed, then loop it.
- **GRAINS trigger out** sends a pulse each time the tape has moved one gap past the main head, a clock at the echo period.
- **Encoder + STOP** (special function) erases the tape: it fades out and a blank tape starts.

TAPE SETTINGS pages replace GRAINS SETTINGS, last after the OTHER SETTINGS pages (also under Menu → Options → 10.TAPE OPTIONS, the last menu pages):

- TAP INPUT, TAPE INPUT (INPUT / LOOP as heard / BOTH), INPUT LEVEL, MONITOR (the swelled input in the tape output)
- GAP MODE, MOTOR, HOLD INPUT, FEEDBACK (MAIN head / ALL heads)
- FOLLOW LOOP PITCH (the tape speed follows the loop pitch), SWELL TIME (GAP / FIXED)

Display (TAPE)

- **Main screen:** the tape moving past the heads, with the record head on the left (its height is the record level). There is one bar per playback head, dotted when it reads backwards.
- **Left bar:** record level, with a line for REGEN.
- **Right bar:** WIDTH.
- **Top bar:** `H<heads>`, or `TP` in tap mode.
- **Bottom bar:**
    - SPEED RANGE (`.5` / `x1` / `x2`), inverted while HOLD is on;
    - the SWELL shape;

    - G = tape speed in semitones.
- **Readout:** turning an overlay knob or flipping a switch shows its name and value for 1.5 s, with `PICKUP` until it reaches a preset's value.
- **GAP SYNC on:** a dotted mark in the loop envelope, one gap behind the play head.

Memory

	GRANULAR	TAPE
Loop memory	360 s (stock)	~285 s
Tape (2 × 2^21 floats)	–	top of the loop array in SDRAM
Tape bridge + engine (~13 KB)	–	heap, created at boot
Grain voice reserve (21 KB heap)	yes	no

DTCM is at the stock level (88.5 %).

Web editor

`firmware/web/` has a Tape section next to Grains, and ENGINE on the card tab. Run its tests with `for t in firmware/web/test/*.test.js; do node "$t"; done`. The sample files are read from `../addac112/support`. One test checks that the editor's `tape_*` keys match what `Preset::save_tape_()` writes.

Not done yet

- Not yet run on hardware. The CPU cost needs measuring on the M7 (on desktop TAPE costs 1.3× to 2.4× CD SKIP), especially with 4 reversed heads at 96 kHz. Above 85 % CPU the bridge already drops to 2 heads.
- Overlay artwork: needs the panel dimensions and knob positions.
- Known engine limit: material played back faster than it was recorded can alias in its top octave. AGE usually removes it first.